



Milk yield dynamics as indicators for disease in dairy cows

B. G. C. de Bruijn,^{1,2*} I. D. E. van Dixhoorn,² B. Kemp,¹ and A. T. M. van Knegsel¹

¹Adaptation Physiology group, Wageningen University & Research, Wageningen, 6708 WD, the Netherlands

²Wageningen Livestock Research, Wageningen University & Research, Wageningen, 6708 WD, the Netherlands

ABSTRACT

Dairy cows experience the majority of diseases during early lactation, and diseases are linked to milk losses. Fluctuations in milk yield were previously suggested to add value in the health monitoring of dairy cows. The aim of this study was to investigate the relations between health status during the first 56 DIM and 2 types of milk yield dynamics: (1) lactation curve characteristics and (2) dynamic indicators of milk yield derived from deviations from the lactation curve. Data on the daily milk yield and disease treatments of 4 previous experiments were merged and examined. Cows were categorized in 3 ways based on disease treatments until 56 DIM: (1) into 3 groups based on the health status: no health problems (NHP), systemic or inflammatory health problems (SHP), other health problems (OHP), (2) by the number of disease treatments per cow (0, 1, or ≥ 2), (3) by the presence or absence of disease treatments for either mastitis, endometritis, claw and leg disorders, vaginal discharge, or cystic ovaries. The following lactation curve characteristics were calculated using the 5-d rolling average of daily milk yield: peak production, day of peak production, slope to peak, total milk yield until 56 DIM, 305-d milk production, and persistency. The following dynamic indicators of milk yield were calculated derived from daily deviations in milk yield during the periods of 10 to 56 DIM and 10 to 305 DIM: log-transformed variance (LnVar), lag-1 autocorrelation (R_{auto}), skewness, and mean negative residuals (MNR). Mixed models were used to investigate relations between disease treatments and milk yield dynamics. Peak production and total milk yield until 56 DIM were lower in SHP cows compared with NHP or OHP cows. Cows of parity ≥ 3 in SHP had a later day of peak production and lower slope to peak production compared with NHP and OHP cows of parity ≥ 3 . Persistency and 305-d production were not affected by health status group. Cows with health problems (SHP and OHP) during the first 56 DIM had a higher LnVar

than cows without treatment for health problems (NHP). More specifically, cows treated for endometritis or cystic ovaries had a higher R_{auto} through lactation compared with healthy cows. The MNR from 10 to 56 DIM was lower for cows treated for claw and leg disorders compared with healthy cows, but MNR was not of added value for other treatments types or health groups, and neither was skewness. The LnVar from 10 to 56 DIM was higher with increasing number of disease treatments. This indicated that cows with health problems in early lactation had more perturbations in milk yield than healthy cows, but the exact response was dependent on the type of disease and number of disease treatments in early lactation. In conclusion, although lactation curve characteristics specifically indicated cows with a systemic or inflammatory health problem, dynamic indicators of milk yield indicated both cows with a systemic or inflammatory health problem and cows with other health problems. The impact of specific diseases on milk yield dynamics should be further investigated to potentially allow for application in on-farm health monitoring tools.
Key words: milk yield, disease, dynamic indicator, lactation curve

INTRODUCTION

Dairy cows experience the majority of diseases during early lactation, with approximately one-third of dairy cows developing at least one clinical disease during this period (Koeck et al., 2012; Ribeiro and Carvalho, 2017). This high incidence of diseases is linked to metabolic and physiological challenges around calving, including recovery from the calving process, onset of lactation, and dietary changes (Goff and Horst, 1997; LeBlanc, 2010; Wankhade et al., 2017). Inability to recover and adapt to these changes increases the risk for diseases, such as mastitis, endometritis, ketosis, or hypocalcemia (Sordillo and Mavangira, 2014; Sundrum, 2015; Aleri et al., 2016). Postpartum diseases impair cow welfare (Sumner et al., 2018) and contribute to economic losses for farmers through milk losses, treatments costs, and increased risk for culling (Carvalho et al., 2019; Pérez-Báez et al., 2021; Puerto et al., 2021).

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*Corresponding author: brigitte.debruijn@wur.nl

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Previous studies have demonstrated relations between diseases and milk production over both short-term (Gröhn et al., 2004; Randall et al., 2016; Venjakob et al., 2018) and long-term periods (De Vliegher et al., 2005; Dubuc et al., 2011; Hostens et al., 2012). For example, clinical mastitis was generally associated with a drop in milk yield on the day of diagnosis and treatment or the preceding days before occurrence of a clinical disease (Gröhn et al., 2004). Heifers with severe sole lesions in the second to fourth month postpartum had an average daily milk yield 2.68 kg lower than heifers without lesions in that period (Randall et al., 2016). Postpartum diseases resulted in long-term effects on the subsequent lactation curve, as the milk yield of healthy cows increased faster to peak production and declined faster after peak production compared with cows that encountered one or more metabolic problems during early lactation (Hostens et al., 2012). In addition, the 305-d milk yield decreased in multiparous cows with both retained placenta and metritis compared with multiparous cows without these diseases during lactation (Dubuc et al., 2011).

Milk yield dynamics have previously been suggested to add value to health monitoring in dairy cows, as perturbation in the lactation curve may reflect how a cow responds to and recovers from challenges (Adriaens et al., 2021a; Ben Abdelkrim et al., 2021). Based on the resilience theory of Scheffer et al. (2018), it was hypothesized that the milk yield of resilient cows is minimally affected by challenges and will quickly recover to the baseline level if it is affected by a challenge (Berghof et al., 2019; Poppe et al., 2021). More resilient cows are therefore better able to cope with diseases, as they are affected by perturbations less frequently and less severely, recover more quickly, and show fewer perturbations in milk yield compared with less resilient cows. Quantifying perturbations in milk yield can help in identifying less resilient cows that have increased risk for diseases, which can support health monitoring approaches.

Resilience indicators derived from daily milk yield records were related to health-related breeding values, such as longevity or fertility (Elgersma et al., 2018; Poppe et al., 2021; Keßler et al., 2025) or clinical mastitis (Kok et al., 2021). The natural log-transformed variance (**LnVar**) and lag-1 autocorrelation (**R_{auto}**) are indicators that can reflect the impact and the length of the impact of certain challenges on the expected lactation curve (Berghof et al., 2019). The LnVar and **R_{auto}** have been associated with breeding values for udder health, fertility, DMI, and longevity in heifers (Poppe et al., 2021). The occurrence of clinical mastitis was reflected by increased LnVar and reduced **R_{auto}** of deviations in milk yield (Kok et al., 2021), but relationships with other diseases, such as endometritis or lameness, or with the frequency of diseases, were not evaluated. In another study, the mean

of negative residuals (**MNR**) from daily step count was indicated to reflect the magnitude of negative deviations in response to a challenge such as a disease occurrence or heat stress event (Poppe et al., 2022). Herewith, it can be hypothesized that similar to the MNR of step counts, the MNR of milk yield is also a suitable resilience indicator in relation with diseases.

To further explore the added value of milk yield dynamics in health monitoring tools, investigating the relations with clinical health data would be valuable, thereby focusing on evaluating different disease types and disease frequency. Hence, the objective of our study was to investigate the relations between (1) lactation curve characteristics until 305 DIM and disease treatments during the first 56 DIM and (2) dynamic indicators of milk yield derived from deviations from the lactation curve during the periods of 10 to 56 DIM and 1 to 305 DIM and disease treatments during the first 56 DIM. The relations of milk yield dynamics with disease treatments until the first 56 DIM were evaluated in 3 ways: (1) 3 groups of cows categorized based on health status (no health problems [**NHP**], systemic or inflammatory health problems [**SHP**], or other health problems [**OHP**]); (2) the number of disease treatments per cow; (3) the presence of treatments for specific diseases including mastitis, endometritis, claw and leg disorders, vaginal discharge, or cystic ovaries. For this study, data from daily milk yield and disease treatments from 4 previously conducted experimental studies were used (van Knegsel et al., 2014; Chen et al., 2015; van Hoeij et al., 2017; Burgers et al., 2021), including data from a total of 579 lactations.

MATERIALS AND METHODS

Experimental Data

This study used data collected during 4 previously published studies, hereafter referred to as study 1a (van Knegsel et al., 2014), study 1b (Chen et al., 2015), study 2 (van Hoeij et al., 2017), and study 3 (Burgers et al., 2021; Table 1). The experimental protocols of all 4 studies were approved by the Institutional Animal Care and Use Committee of Wageningen University & Research (the Netherlands) and comply with the Dutch law on Animal Experimentation (study 1a and 1b: protocol number 2010026; study 2: protocol number 2014125; study 3: protocol number 2016.D-0038.005).

All 4 studies included Holstein Friesian dairy cows at the Dairy Campus research herd (Wageningen Livestock Research, Lelystad [study 1a, 1b, 2] or Leeuwarden [study 3], the Netherlands). In study 1a, 168 cows were randomly assigned to 1 of 3 dry period lengths (**DPL**): 0 d, 30 d, or 60 d. After one full lactation in study 1a, 130 of the 168 cows were included in study 1b during the

Table 1. Number of lactations and experimental treatments of the included studies¹

Item	Study 1a: van Kneegsel et al. (2014)	Study 1b: Chen et al. (2015)	Study 2: van Hoeij et al. (2017)	Study 3: Burgers et al. (2021)
No. of cows	168	130	127	154
Treatment	0, 30 or 60-d DPL	0, 30 or 60-d DPL	0 or 30-d DPL	50, 125 or 200-d VWP

¹DPL = dry period length, VWP = voluntary waiting period.

next lactation and remained in their DPL group. Nineteen of the cows originally assigned to the 0-d DPL were subjected to a 67-d DPL in study 1b (actual DPL: 67 ± 8 d). Cows were allocated to this new group and dried off when they had a milk yield of <4 kg/d at least 30 d before the expected calving date (Chen et al., 2015). In study 2, a total of 127 cows were randomly assigned to 1 of 2 DPL: 0 d or 30 d. In study 3, a total of 154 cows were randomly assigned to a voluntary waiting period (VWP) until first insemination of 50, 125, or 200 d. Daily milk yield data and records of disease treatments from the complete lactation from the 4 studies were merged and used in this research.

Housing and Diet

In all studies, cows were housed in a freestall with cubicles and a slatted floor. Cows were milked twice daily (~0600 h and ~1800 h). Ration and feeding strategies were described earlier (study 1a/1b in van Kneegsel et al., 2014; study 2 in van Hoeij et al., 2017; study 3 in Burgers et al., 2021). In short, in study 1a and 1b lactating cows were fed a basal mixed ration which supported 25 kg of milk yield per day and consisted of grass silage, corn silage, wheat straw, and rapeseed meal or soybean meal (51:34:2:13, DM basis). The lactating cows in study 2 were fed a basal mixed ration supporting 25 kg of milk yield per day and consisted of grass silage, corn silage, wheat straw, soybean meal, and sugar beet pulp (44:34:2:10:8, DM basis). In study 3, a basal mixed ration was fed which supported 22 kg of milk yield per day and consisted of grass silage, corn silage, soybean meal, and wheat meal (60:20:11:9, DM basis). From d 10 before the expected calving date, both lactating and dry cows were fed 1 kg/d of concentrate. After calving, concentrate supply increased stepwise until 8.5 kg of milk/d (1a and 1b), until 6.7 kg of milk/d (study 2), or until 9 kg of milk/d for primiparous cows and 10 kg of milk/d for multiparous cows (study 3). In all studies, 1 kg of additional concentrate was supplied daily in the milking parlor, and cows had free access to water and the basal mixed ration.

Measurements. Milk yield was recorded at every milking (twice daily) from the day of calving until the dry-off date. All disease treatments during the experiment were recorded by trained farm staff according to

farm protocols (Dairy Campus, Lelystad [study 1a, 1b, 2] or Leeuwarden [study 3]). Disease treatments were registered daily based on treatments for disease, including hypocalcemia, retained placenta, endometritis, cystic ovaries, vaginal discharge, mastitis, claw and leg disorders, fever, displaced abomasum, intestine problems, and other diseases (peritonitis, pneumonia, teat stepping). Disease treatment data were filtered based on the following principle as also reported by Wang et al. (2024): if the same disease treatment took place more than once within 10 d for the same cow, only the first disease treatment remained in the dataset. If the interval between similar disease treatments was larger than 10 d, the disease case was treated as a separate disease case. Claw or leg treatments were treated as one disease occurrence when the interval between treatments was ≤ 28 d. An overview of disease treatments, milk yield, and fat- and protein-corrected milk yield per study is presented in Table 2.

Calculation of Lactation Curve Characteristics. To calculate lactation curve characteristics, first, milk yield records were summed per cow per day to obtain daily milk yield records (kg of milk/d; $n = 199,451$). Second, a 5-d rolling average of milk yield was calculated for each lactation curve and used for further calculations of lactation curve characteristics. Peak production (kg of milk/d), day of peak production, and slope from 10 DIM to peak production (kg of milk/d) were calculated based on the approach of Burgers et al. (2023). Lactation persistency was defined as the reduction in milk yield after peak yield, and was calculated between 100 and 260 DIM (Burgers et al., 2021). Finally, total milk yield until 56 DIM (kg) and 305-d production (kg) were calculated.

Calculation of Dynamic Indicators of Milk Yield. To calculate dynamic indicators of milk yield, first, milk yield records were summed per cow per day to obtain daily milk yield records, as described previously. Second, individual lactation curves were fitted as by Poppe et al. (2020) to estimate the expected milk yield per cow per day based on daily milk yield records:

$$y_t = \beta_0 + \beta_1 \times t + \beta_2 \times t^2 + \beta_3 \times t^3 + \beta_4 \times t^4 + \varepsilon,$$

where y_t is the observed milk yield on DIM t ; t^n are DIM to the power of n , where n is 1, 2, 3, or 4; β_n are regression coefficients describing the relationships between t^n and y_t ; and ε is the error term. A quantile of 0.7 was

Table 2. Disease treatments, milk yield (kg of milk/d) and fat- and protein-corrected milk yield (FPCM; kg of milk/d) until 305 DIM per study,¹ and number and % (in parentheses) of disease treatments within 56 DIM for each disease treatment type (treatments by 56 DIM)

Item	Study				Treatments by 56 DIM
	1a	1b	2	3	
No. of cows	161	119	124	115	
Systemic and inflammatory treatments					
Mastitis	61	95	47	47	56 (22.4)
Endometritis	34	27	17	9	64 (73.5)
Retained placenta	16	10	9	4	39 (100)
Fever	13	11	9	3	32 (88.9)
Hypocalcemia	5	11	5	1	22 (100)
Displaced abomasum	1	2	2	0	5 (100)
Other treatments					
Claw and leg disorders	235	150	76	92	61 (11.0)
Vaginal discharge	40	26	19	34	67 (56.3)
Cystic ovaries	36	57	40	37	60 (35.2)
Intestine problems	1	5	0	0	6 (100)
Other	12	12	5	7	21 (58.3)
Total no. of disease treatments	454	406	229	234	433
Total no. of cows with disease treatments	148	110	93	90	297
Milk yield ² (kg of milk/d)	27.7 (11.9)	29.6 (11.6)	26.9 (9.15)	28.6 (9.76)	
FPCM ² (kg of milk/d)	28.9 (11.8)	31.2 (11.1)	28.4 (9.46)	30.9 (8.58)	

¹The numbers of cows within each study treated for a disease are shown.

²Average daily yield until 305 DIM is presented as mean (SD).

used to approximate the lactation curve in absence of disturbances in line with Poppe et al. (2020). Curves were fitted using the “quantreg” package (Koenker, 2009) and the poly function in R4.4.1 (<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/poly>).

After fitting the individual lactation curves, the deviations from the expected daily milk yield were calculated as $y_t - \hat{y}_t$ for each individual lactation curve. The deviations for 1 to 10 DIM were removed because of poor fit of the lactation curve model. All milk yield records after 305 DIM were removed to standardize lactation length for all cows. Lactations with >5% missing daily milk yield records were discarded, resulting in a final dataset of 176,827 daily milk records of 519 lactations of 385 distinct cows.

Three dynamic indicators of milk yield were calculated for 2 periods (10–56 DIM and 10–305 DIM) based on the deviations from the expected milk yield: LnVar, R_{auto} , skewness and the MNR of milk yield. The indicators of individual cows were set to missing if they deviated by more than 4 SD from the mean of that indicator in the periods of 10 to 56 or 10 to 305 DIM (<0.001% of the data).

Categorizing Cows Based on Disease Treatments

Cows were categorized in 3 different ways based on disease treatments received up to 56 DIM: (1) based on health status group, (2) based on the number of disease treatments per cow, and (3) based on the presence of disease treatments for the 5 disease types with the highest incidence.

For the categorization based on health status groups (adapted from Hostens et al., 2012), cows were categorized in one of 3 groups: cows without any disease treatments during the first 56 DIM (NHP); cows with one or more systemic or inflammatory disease treatment during the first 56 DIM (SHP); or cows with other disease treatments during the first 56 DIM (OHP). Treatments for endometritis, fever, hypocalcemia, mastitis, retained placenta, or displaced abomasum were defined as systemic and inflammatory diseases. Treatment for cystic ovaries, vaginal discharge, claw and leg problems, intestine problems, or other diseases were defined as other health problems. Health status groups were adapted from Hostens et al. (2012) with the adaptation in the current study that the group with no systemic and inflammatory health treatments was split into 2 groups: OHP and NHP. In addition, the metabolic disease category (Hostens et al., 2012) was renamed to systemic and inflammatory health problems, and fever was added to this category.

For categorization based on the presence or absence of a specific disease treatment, the treatment types with the highest incidence until 56 DIM were used: mastitis, endometritis, lameness, vaginal discharge, and cystic ovaries. Other treatments were not evaluated due to low incidence in the dataset. Cows were assigned to 1 of 3 groups per treatment type until 56 DIM: (1) disease treatment for the specific disease was recorded, (2) a disease treatment was recorded, but not for the specific disease, and (3) no disease treatment was recorded (corresponds with NHP). Per specific disease, only cows in groups 1 and 3 were considered in the further analysis.

Table 3. Number of cows per parity class (1 and 2, ≥ 3), per health status group,¹ and per number of disease treatments during the first 56 DIM (0, 1, ≥ 2)

Parity class	Health status group			Number of disease treatments		
	NHP	SHP	OHP	0	1	≥ 2
1 and 2	108	36	38	108	49	25
≥ 3	114	139	84	114	131	92
Total	222	175	122	222	180	117

¹NHP = no health problems, SHP = systemic or inflammatory health problems, and OHP = other health problems, during first 56 DIM.

For categorization based on the number of disease treatments per cow until 56 DIM, cows were assigned to one of 3 groups: 0, 1, or ≥ 2 disease treatments until 56 DIM. An overview of the number of cows per health status group and number of cows per treatment number per parity class can be found in Table 3.

Statistical Analysis

All statistical analyses were performed using SAS (version 9.4, SAS Institute Inc., Cary, NC). To evaluate normality of residuals, a normality test (PROC UNIVARIATE in SAS) was performed where skewness between -1 and 1 and kurtosis between -2 and 2 were used as criteria for normality. Tukey's studentized range tests were performed to investigate between-group differences, where $P < 0.05$ was considered a significant difference. Values were shown as LSM and the largest SEM of the presented groups.

To evaluate relations between health status from 10 to 56 DIM and milk yield dynamics, a mixed model (PROC MIXED in SAS) was used. The dependent variable was one of the lactation curve characteristics or one of the dynamic indicators of milk yield either from 10 to 56 DIM or 10 to 305 DIM. Independent variables were health status (NHP, SHP, or OHP), parity class (parity 1 and 2, or parity ≥ 3), study (1a, 1b, 2, or 3), and their 2-way interactions. Study treatment was included as random effect (0, 30, or 60-d DPL or 50, 125, 200-d VWP). Preliminary analyses showed that the different diets in study 1a, 1b, and 2 had no added value to the models, and was therefore not included in any of the further analyses. The 2-way interactions between study and parity class could not be tested because not all combinations were present in the data, and this interaction was not included in any further analyses. In addition, the 2-way interactions between health status and study was never significant for any of the dependent variables, and was therefore excluded from the model ($P < 0.05$).

To evaluate relations between the number of disease treatments per cow until 56 DIM and milk yield

dynamics, a mixed model (PROC MIXED) was used. The dependent variable was one of the lactation curve characteristics or one of the dynamic indicators of milk yield either from 10 to 56 DIM or 10 to 305 DIM. Independent variables were number of disease treatments per cow until 56 DIM (0, 1, or ≥ 2), parity class (parity 1 and 2, or parity ≥ 3), study (1a, 1b, 2, or 3), and the 2-way interaction between disease treatment and parity class. Study treatment was included as random effect (0, 30 or 60-d DPL or 50, 125, 200-d VWP). The 2-way interaction between number of disease treatments and study was never significant for any of the dependent variables and was therefore excluded from the model ($P < 0.05$).

To evaluate relations between the 5 disease types with the highest disease incidence until 56 DIM and milk yield dynamics, a mixed model (PROC MIXED in SAS) was used. The following disease treatments during the first 56 DIM were evaluated separately: mastitis, endometritis, claw and leg disorders, vaginal discharge, and cystic ovaries. The dependent variable was one of the lactation curve characteristics or one of the dynamic indicators of milk yield either from 10 to 56 DIM or 10 to 305 DIM. Independent variables were disease treatment for one of the specific diseases (yes or no disease treatment at all), parity class (parity 1 and 2, or parity ≥ 3), study (1a, 1b, 2, or 3), and the 2-way interaction between disease treatment and parity class. Study treatment was included as random effect (0, 30 or 60-d DPL or 50, 125, 200-d VWP). The 2-way interactions between disease treatment and study could not be tested because not all combinations were present in the data.

RESULTS

Milk Yield Dynamics and Health Status Group

Lactation curves and dynamic indicators of milk yield were investigated per health status (Table 4). Peak production and total milk yield until 56 DIM were related to health status group ($P \leq 0.01$; Figure 1), where cows in SHP had a lower peak production and total milk yield until 56 DIM compared with cows in NHP or OHP. An interaction between health status group and parity class was present for day of peak production and slope to peak production ($P = 0.02$; Figure 2A–B). Day of peak production and slope to peak production differed between health status groups for cows of parity ≥ 3 , but not for cows of parity 1 and 2. For parity ≥ 3 , day of peak production was highest for SHP cows, lowest for NHP cows, with OHP cows in between. Slope to peak production was lower for SHP cows compared with NHP and OHP cows of parity ≥ 3 .

The LnVar from 10 to 56 DIM was related to health status ($P = 0.04$), where NHP cows had a lower LnVar

Table 4. Lactation curve characteristics and dynamic indicators of milk yield for cows with no disease treatments (NHP), treatments for systemic or inflammatory disease (SHP), or other health treatments (OHP) during the first 56 DIM¹

Item ^{3,4,5,6}	Health status group			SEM	P-value ²			
	NHP	SHP	OHP		HS	P	S	HS × P
No. cows	222	175	122					
Peak production (kg of milk/d)	42.9 ^b	40.7 ^a	42.4 ^b	1.33	0.01	<0.01	0.40	0.30
Day of peak production	41.7	45.6	42.1	1.69	0.04	<0.01	0.01	0.02
Slope to peak (kg of milk/d)	0.30	0.28	0.32	0.02	0.73	<0.01	0.02	0.02
Total milk yield until 56 DIM (kg)	2060 ^b	1904 ^a	2027 ^b	66.9	<0.01	<0.01	0.07	0.22
305-d production (kg)	9,076	8,906	9,019	292	0.68	<0.01	0.16	0.28
Persistency (kg of milk/d)	-0.09	-0.09	-0.09	0.003	0.16	<0.01	<0.01	0.82
LnVar 10–56	1.43 ^a	1.60 ^b	1.61 ^b	0.08	0.04	<0.01	0.03	0.96
LnVar 10–305	1.34	1.38	1.39	0.08	0.73	<0.01	<0.01	0.56
R _{auto} 10–56	0.40	0.46	0.46	0.03	0.10	0.97	0.03	0.14
R _{auto} 10–305	0.52	0.54	0.55	0.02	0.25	0.11	<0.01	0.41
Skewness 10–56	-0.47	-0.55	-0.41	0.06	0.33	0.16	<0.01	0.04
Skewness 10–305	-0.56	-0.75	-0.67	0.10	0.20	0.67	<0.01	0.32
MNR 10–56	-1.50	-1.63	-1.66	0.11	0.15	<0.01	0.04	0.53
MNR 10–305	-1.51	-1.51	-1.53	0.11	0.92	<0.01	<0.01	0.41

^{a,b}Values within the same row with different superscripts significantly differ ($P < 0.05$).

¹Values are represented as least squares means $\pm$ maximum SEM.

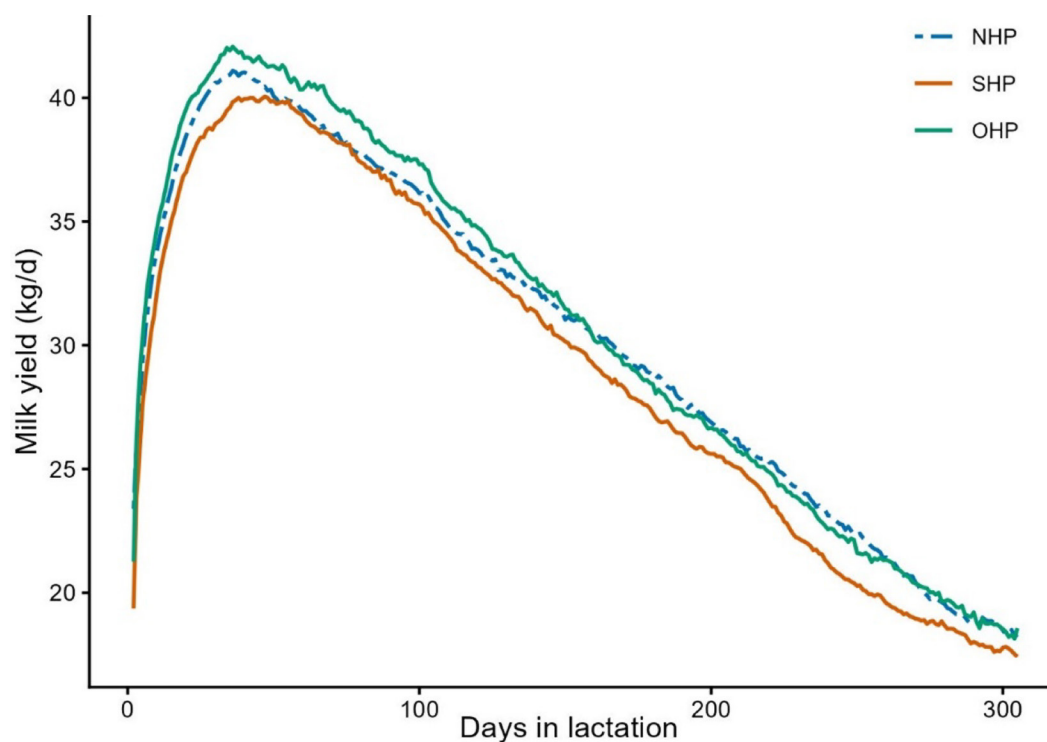
²HS = health status group, P = parity class (1+2, ≥ 3), S = study (1a, 1b, 2, 3).

³Day of peak production was log-transformed, and slope to peak was transformed by a square root to obtain normally distributed residuals. Back-transformed values are presented here.

⁴Persistency was calculated between 100 and 260 DIM.

⁵LnVar = natural log-transformed variance, R_{auto} = lag-1 autocorrelation, and MNR = mean negative residuals of milk yield, calculated over the periods of 10–56 DIM and 10–305 DIM.

⁶Skewness and MNR were transformed by a natural logarithm to obtain normally distributed residuals. Back-transformed values are presented here.

**Figure 1.** Lactation curves (kg of milk/d) until 305 DIM per health status group (no health problems [NHP], systemic or inflammatory health problems [SHP], or other health problems [OHP], until 56 DIM).

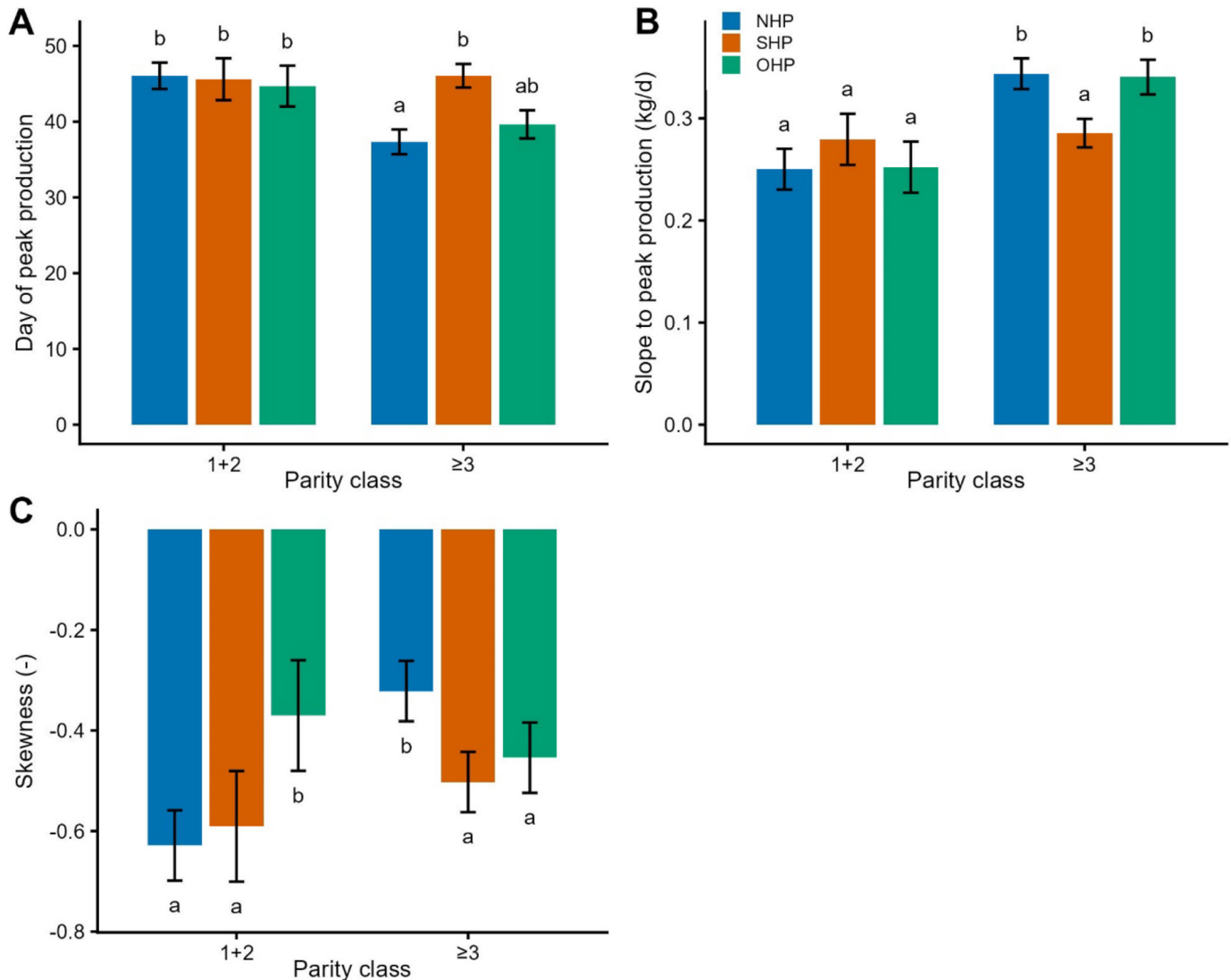


Figure 2. Interaction of health status group (no health problems [NHP], systemic or inflammatory health problems [SHP], or other health problems [OHP], during the first 56 DIM) and parity class (1+2, ≥ 3) for (A) day of peak production and (B) slope to peak production (kg of milk/d) and (C) skewness from 10 to 56 DIM. Values represent LSM $\pm$ SE. Values with different letters (a,b) differ ($P < 0.05$).

from 10 to 56 DIM compared with SHP and OHP cows. The LnVar and MNR from 10 to 56 DIM and 10 to 305 DIM were related to parity class ($P < 0.01$), where from 10 to 56 DIM and 10 to 305 DIM, cows of parity 1 and 2 had a lower LnVar (10–56 DIM: 1.37 vs. 1.73, 10–305 DIM: 1.18 vs. 1.56) and higher MNR (10–56 DIM: -1.47 vs. -1.74 , 10–305 DIM: -1.40 vs. -1.64) compared with cows of parity ≥ 3 .

Skewness from 10 to 56 DIM was affected by an interaction between health status group and parity class ($P = 0.04$; Figure 2C). In parity 1 and 2, OHP cows had a less negative skewness compared with NHP and SHP cows. In parity ≥ 3 , NHP cows had a less negative skewness compared with SHP and OHP cows. Dynamic indicators

of milk yield calculated for the period of 10 to 305 DIM were not related to health status or the interaction with parity class.

Milk Yield Dynamics and Number of Disease Treatments

Lactation curves characteristics and dynamic indicators of milk yield were investigated for the number of disease treatments per cow until 56 DIM (Table 5). Total milk yield until 56 DIM was related to number of disease treatments ($P < 0.01$; Figure 3), where cows with 1 or ≥ 2 disease treatments until 56 DIM had a lower total milk yield until 56 DIM compared with cows without any

Table 5. Lactation curve characteristics and dynamic indicators of milk yield for cows with 0, 1 or ≥ 2 disease treatment during the first 56 DIM¹

Item ^{3,4,5,6}	Number of treatments			SEM	P-value ²			
	0	1	≥ 2		NT	P	S	NT $\times$ P
No. cows	222	180	117					
Peak production (kg of milk/d)	42.9	41.4	41.2	1.39	0.05	<0.01	0.41	0.56
Day of peak production	41.3	44.2	44.4	1.9	0.17	0.03	0.01	0.01
Slope to peak (kg of milk/d)	0.29	0.28	0.29	0.03	0.64	<0.01	0.05	0.15
Total milk yield until 56 DIM (kg)	2,059 ^b	1,978 ^a	1,915 ^a	69.9	<0.01	<0.01	0.07	0.29
305-d production (kg)	9,069	8,969	8,893	306	0.68	<0.01	0.14	0.88
Persistency (kg of milk/d)	-0.09	-0.09	-0.09	0.004	0.45	<0.01	<0.01	0.48
LnVar 10–56	1.43 ^a	1.55 ^{ab}	1.69 ^b	0.10	0.02	<0.01	0.04	0.98
LnVar 10–305	1.34	1.34	1.46	0.09	0.26	<0.01	<0.01	0.59
R _{auto} 10–56	0.40	0.44	0.48	0.03	0.07	0.67	0.04	0.33
R _{auto} 10–305	0.52	0.53	0.57	0.02	0.10	0.16	<0.01	0.50
Skewness 10–56	-0.47	-0.56	-0.56	0.09	0.42	0.11	<0.01	0.02
Skewness 10–305	-0.56	-0.69	-0.72	0.12	0.24	0.69	<0.01	0.39
MNR 10–56	-1.50	-1.59	-1.74	0.11	0.08	<0.01	0.04	0.61
MNR 10–305	-1.51	-1.49	-1.57	0.08	0.48	<0.01	<0.01	0.47

^{a,b}Values within the same row with different superscripts significantly differ ($P < 0.05$).

¹Values are represented as LSM $\pm$ maximum SEM.

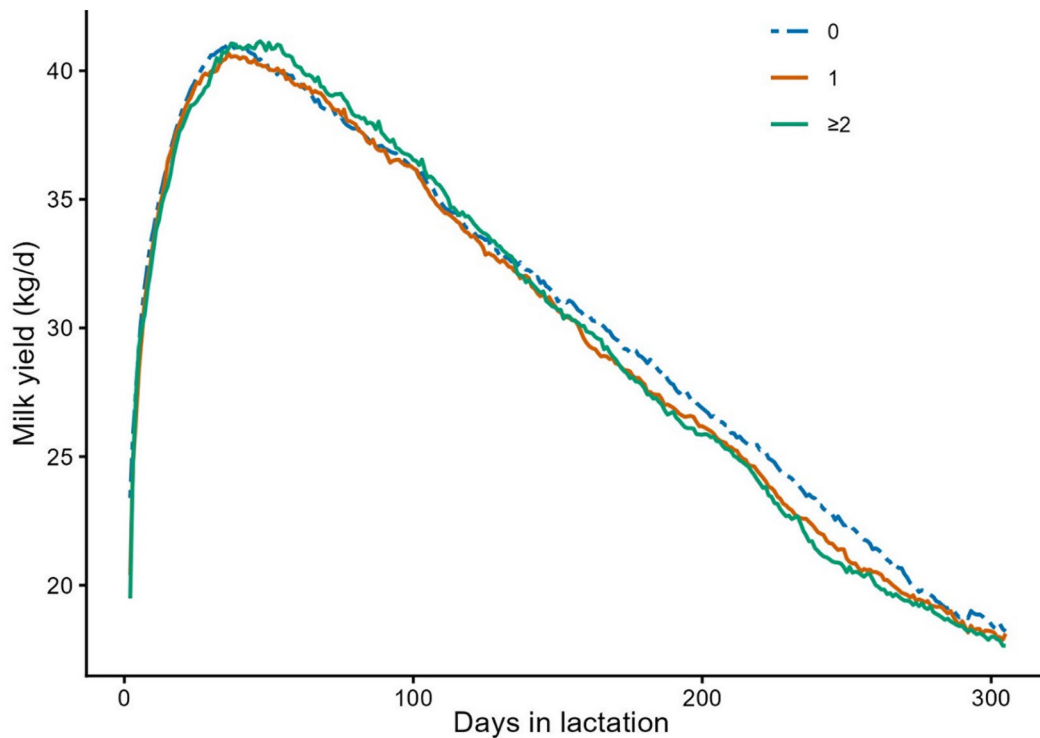
²NT = number of disease treatments, P = parity class (1+2, ≥ 3), S = study (1a, 1b, 2, 3).

³Day of peak production was log-transformed and slope to peak was transformed by a square root to obtain normally distributed residuals. Back-transformed values are presented here.

⁴Persistency was calculated between 100 and 260 DIM.

⁵LnVar = natural log-transformed variance, R_{auto} = lag-1 autocorrelation, and MNR = mean negative residuals of milk yield, calculated over the period 10–56 DIM and 10–305 DIM.

⁶Skewness and MNR were transformed by a natural logarithm to obtain normally distributed residuals. Back-transformed values are presented here.

**Figure 3.** Lactation curves (kg of milk/d) until 305 DIM for cows with 0, 1, or ≥ 2 disease treatments until 56 DIM.

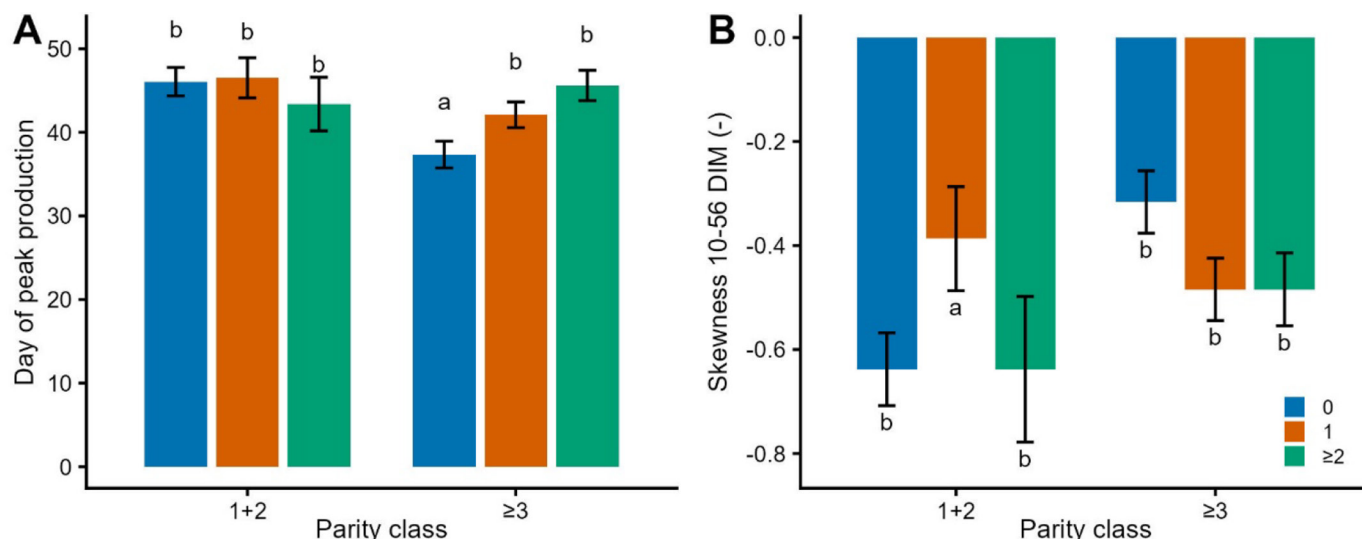


Figure 4. Interaction of number of disease treatment until 56 DIM (0, 1, or ≥ 2) and parity class (1+2, ≥ 3) for (A) day of peak production and (B) skewness from 10 to 56 DIM. Values represent LSM $\pm$ SE. Values with different letters (a,b) differ ($P < 0.05$).

disease treatments in this period. The relation between day of peak production and number of disease treatments depended on parity class ($P < 0.01$; Figure 4A), where cows of parity ≥ 3 with 1 or ≥ 2 disease treatments until 56 DIM had a later day of peak production compared with cows without any disease treatments in this period. Day of peak production was not related to number of disease treatments for parity 1 and 2. No other lactation curve characteristics were related to the number of disease treatments or an interaction with parity class.

The LnVar from 10 to 56 DIM was related to the number of disease treatments until 56 DIM ($P = 0.02$), where cows with ≥ 2 disease treatments had the highest LnVar, followed by cows with 1 disease treatment, and cows without any disease treatment had the lowest LnVar from 10 to 56 DIM. An interaction between number of disease treatments and parity class was found for skewness from 10 to 56 DIM (Figure 4B), where cows of parity 1 and 2 with 1 disease treatment until 56 DIM had a lower skewness compared with cows with 0 or ≥ 2 disease treatment in this period. The LnVar of cows of parity ≥ 3 did not differ between different number of disease treatments. No other dynamic indicators of milk yield were related to the number of disease treatments or an interaction with parity class.

Milk Yield Dynamics for 5 Treatment Types

Lactation curves characteristics and dynamic indicators of milk yield were investigated for 5 diseases treatment types with the highest incidence until 56 DIM (Tables 6 and 7). Peak production, total milk yield until 56 DIM, and 305-d production ($P < 0.01$) were lower

for cows treated for mastitis treatment during the first 56 DIM compared with cows without any disease treatments in this period (Figure 5). Peak production and total milk yield during the first 56 DIM ($P < 0.01$) were lower for cows with endometritis treatment compared with cows without any disease treatments in this period.

The relations between day of peak production and treatment for mastitis ($P = 0.01$), endometritis ($P = 0.03$), claw and leg disorders ($P < 0.01$), and vaginal discharge ($P = 0.03$; Figure 6) were all affected by parity class. Cows of parity ≥ 3 treated for mastitis, endometritis, claw and leg disorders, or vaginal discharge during the first 56 DIM had a later day of peak production compared with cows of parity ≥ 3 without any treatment in this period. Cows of parity 1 and 2 with and without treatment had a similar day of peak production. In addition, the relation between slope to peak production and mastitis treatment was affected by parity class ($P < 0.05$), where cows of parity ≥ 3 treated for mastitis during the first 56 DIM had a lower slope to peak production compared with cows of parity ≥ 3 without any treatment in this period. The slope to peak production did not differ for cows of parity 1 and 2 with or without a mastitis treatment. Lactation curve characteristics were not related to treatment for cystic ovaries.

Cows with treatment for claw and leg disorders had a higher LnVar from 10 to 56 DIM ($P < 0.01$; Table 7), and cows treated for cystic ovaries had a higher LnVar from 10 to 305 DIM ($P < 0.01$) compared with cows without any disease treatment during the first 56 DIM. LnVar from 10 to 305 DIM was related to an interaction between parity class and treatment for vaginal discharge during the first 56 DIM ($P = 0.04$; Figure 7A), where cows of parity ≥ 3

Table 6. Lactation curve characteristics for cows without any disease treatment (n = 222) or with a specific disease treatment during the first 56 DIM¹

Disease treatment type	Lactation curve characteristic ³	Disease treatment			P-value ²			
		Yes	No	SEM	DT	P	S	DT × P
Mastitis (n = 56)	Peak production (kg of milk/d)	37.6 ^a	42.7 ^b	1.90	<0.01	<0.01	0.40	0.69
	Day of peak production	41.6	38.9	1.08	0.42	0.93	0.09	0.01
	Slope to peak (kg of milk/d)	0.29	0.30	0.03	0.93	0.38	0.05	0.05
	Total MY first 56 DIM (kg)	1,763 ^a	2,054 ^b	97.6	<0.01	<0.01	0.07	0.95
	305-d production (kg)	7,845 ^a	9,021 ^b	470	<0.01	<0.01	0.07	0.6
Endometritis (n = 64)	Persistency (kg of milk/d)	-0.08	-0.09	0.008	0.16	<0.01	<0.01	0.86
	Peak production (kg of milk/d)	40.5 ^a	42.8 ^b	1.37	<0.01	<0.01	0.46	0.51
	Day of peak production	46.8 ^b	41.3 ^a	1.00	0.01	0.12	0.04	0.03
	Slope to peak (kg of milk/d)	0.28	0.30	0.03	0.42	<0.01	0.03	0.21
	Total MY first 56 DIM (kg)	1,894 ^a	2,062 ^b	66.7	<0.01	<0.01	0.07	0.39
Claw and leg disorders (n = 61)	305-d production (kg)	8,742	9,062	312	0.22	<0.01	0.15	0.49
	Persistency (kg of milk/d)	-0.09	-0.09	0.005	0.19	<0.01	<0.01	0.96
	Peak production (kg of milk/d)	42.0	42.7	1.57	0.47	<0.01	0.69	0.95
	Day of peak production	42.5	41.4	1.10	0.66	0.67	0.06	<0.01
	Slope to peak (kg of milk/d)	0.28	0.30	0.03	0.46	0.03	<0.01	0.22
Vaginal discharge (n = 76)	Total MY first 56 DIM (kg)	1,971	2,052	79.0	0.12	<0.01	0.13	0.73
	305-d production (kg)	8,642	9,021	343	0.14	<0.01	0.28	0.69
	Persistency (kg of milk/d)	-0.09	-0.09	0.005	0.87	<0.01	<0.01	0.87
	Peak production (kg of milk/d)	41.4	42.6	1.42	0.15	<0.01	0.41	0.09
	Day of peak production	44.7	41.6	2.13	0.12	0.05	0.25	0.03
Cystic ovaries (n = 60)	Slope to peak (kg of milk/d)	0.27	0.30	0.03	0.17	<0.01	0.05	0.92
	Total MY first 56 DIM (kg)	1,966	2,046	74.5	0.08	<0.01	0.06	0.32
	305-d production (kg)	8,911	9,019	299	0.63	<0.01	0.06	0.18
	Persistency (kg of milk/d)	-0.09	-0.09	0.004	0.90	<0.01	0.01	0.47
	Peak production (kg of milk/d)	43.9	42.7	1.38	0.22	<0.01	0.61	0.96
	Day of peak production	42.2	41.5	2.47	0.76	0.03	0.17	0.24
	Slope to peak (kg of milk/d)	0.32	0.29	0.02	0.21	<0.01	0.12	0.40
	Total MY first 56 DIM (kg)	2,107	2,052	71.0	0.29	<0.01	0.10	0.34
	305-d production (kg)	9,550	9,031	304	0.05	<0.01	0.10	0.76
	Persistency (kg of milk/d)	-0.09	-0.09	0.005	0.27	<0.01	<0.01	0.41

^{a,b}Values within the same row with different superscripts significantly differ ($P < 0.05$).

¹Values are represented as LSM ± maximum SEM.

²DT = disease treatment (yes, no), P = parity class (1+2, ≥3), S = study (1a, 1b, 2, 3).

³Day of peak production was log-transformed and slope to peak was transformed by a square root to obtain normally distributed residuals; back-transformed values are presented here. Total MY first 56 DIM = total milk yield during the first 56 DIM. Persistency was calculated between 100 and 260 DIM.

had a higher LnVar from 10 to 305 DIM compared with cows of parity 1 and 2. Cows treated for endometritis or cystic ovaries had a higher R_{auto} from 10 to 56 DIM and 10 to 305 DIM ($P < 0.05$) compared with cows without any disease treatment during this period. Cows treated for claw or leg disorders had a higher R_{auto} from 10 to 305 DIM ($P = 0.02$) compared with cows without any disease treatment during the first 56 DIM. Cows treated for a claw or leg disorder during the first 56 DIM had a lower MNR from 10 to 305 ($P = 0.04$) compared with cows without any disease treatments during this period.

Cows treated for endometritis had a lower skewness from 10 to 305 DIM ($P = 0.03$) compared with cows without any disease treatment during the first 56 DIM. Skewness often interacted with disease treatment and parity class (Figure 7B–E). Cows of parity ≥3 that were treated for claw and leg disorders had a lower skewness compared with cows of parity ≥3 without any disease treatments during this period ($P < 0.05$). Cows of parity

1 and 2 that were treated for cystic ovaries had a higher skewness compared with cows of parity 1 and 2 without any disease treatments during this period. Cows of parity ≥3 that were treated for mastitis or vaginal discharge had a lower skewness from 10 to 305 DIM compared with cows of parity ≥3 without any disease treatments during this period ($P \leq 0.04$), although there was no difference in skewness between cows of parity 1 and 2 with or without disease treatment.

DISCUSSION

In this study, lactation curve characteristics, describing the shape of the lactation curve, and dynamic indicators of milk yield, indicating the perturbations in milk yield during different stages of the lactation, were evaluated for cows with different types and numbers of disease treatments during the first 56 DIM. Both the shape of the lactation curve and perturbations in milk yield were

Table 7. Dynamic indicators of milk yield derived from deviations from individual lactation curves for cows without any disease treatment (n = 222) or with specific disease treatment during the first 56 DIM¹

Treatment type	Item ³	Period (DIM)	Disease treatment			P-value ²			
			Yes	No	SEM	DT	P	S	DT × P
Mastitis (n = 56)	LnVar	10–56	1.53	1.42	0.07	0.53	<0.01	0.06	0.07
		10–305	1.23	1.33	0.17	0.56	<0.01	<0.01	0.12
	R _{auto}	10–56	0.46	0.40	0.06	0.30	0.69	0.23	0.96
		10–305	0.53	0.51	0.05	0.78	0.49	<0.01	0.59
	Skewness	10–56	–0.80	–0.48	0.07	0.07	0.13	<0.01	0.75
		10–305	–0.51	–0.59	0.09	0.79	0.14	<0.01	0.04
Endometritis (n = 64)	MNR	10–56	–1.72	–1.49	0.10	0.22	0.02	0.07	0.17
		10–305	–1.34	–1.51	0.09	0.18	<0.01	<0.01	0.06
	LnVar	10–56	1.57	1.44	0.10	0.16	0.03	0.10	0.19
		10–305	1.35	1.34	0.10	0.98	<0.01	<0.01	0.21
	R _{auto}	10–56	0.48 ^b	0.40 ^a	0.04	0.04	0.51	0.57	0.93
		10–305	0.57 ^b	0.51 ^a	0.03	0.05	0.08	<0.01	0.16
Claw and leg disorders (n = 61)	Skewness	10–56	–0.53	–0.46	0.04	0.52	0.05	<0.01	0.38
		10–305	–0.89 ^a	–0.60 ^b	0.05	0.03	0.45	<0.01	0.77
	MNR	10–56	–1.62	–1.51	0.11	0.29	0.45	0.08	0.49
		10–305	–1.49	–1.51	0.11	0.70	<0.01	<0.01	0.22
	LnVar	10–56	1.74 ^b	1.44 ^a	0.12	<0.01	<0.01	0.06	0.62
		10–305	1.46	1.34	0.10	0.20	<0.01	<0.01	0.95
Vaginal discharge (n = 76)	R _{auto}	10–56	0.46	0.40	0.04	0.14	0.34	0.60	0.12
		10–305	0.58 ^b	0.51 ^a	0.03	0.02	0.82	<0.01	0.67
	Skewness	10–56	–0.46	–0.46	0.04	0.96	0.81	<0.01	<0.01
		10–305	–0.72	–0.56	0.05	0.29	0.94	<0.01	0.32
	MNR	10–56	–1.77 ^a	–1.51 ^b	0.11	0.04	<0.01	0.04	0.17
		10–305	–1.58	–1.51	0.10	0.29	0.02	<0.01	0.84
Cystic ovaries (n = 60)	LnVar	10–56	1.44	1.44	0.08	0.98	<0.01	<0.01	0.79
		10–305	1.28	1.34	0.08	0.43	<0.01	<0.01	0.04
	R _{auto}	10–56	0.39	0.40	0.03	0.73	0.79	0.40	0.34
		10–305	0.51	0.51	0.03	0.88	0.11	<0.01	0.16
	Skewness	10–56	–0.53	–0.48	0.04	0.54	0.04	<0.01	0.16
		10–305	–0.72	–0.59	0.05	0.17	0.15	<0.01	<0.01
	MNR	10–56	–1.54	–1.51	0.10	0.76	<0.01	<0.01	0.39
		10–305	–1.46	–1.51	0.11	0.42	<0.01	<0.01	0.05
	LnVar	10–56	1.58	1.43	0.10	0.15	<0.01	0.02	0.22
		10–305	1.53 ^b	1.34 ^a	0.10	0.04	<0.01	<0.01	0.92
	R _{auto}	10–56	0.50 ^b	0.40 ^a	0.04	0.01	0.76	0.28	0.75
		10–305	0.59 ^b	0.51 ^a	0.03	0.01	0.84	<0.01	0.44
	Skewness	10–56	–0.29	–0.48	0.04	0.06	0.37	<0.01	0.05
		10–305	–0.66	–0.59	0.05	0.50	0.04	<0.01	0.26
	MNR	10–56	–0.21	–0.22	0.09	0.88	<0.01	<0.01	0.19
		10–305	–0.23	–0.22	0.09	0.19	<0.01	<0.01	0.60

^{a,b}Values within the same row with different superscripts significantly differ ($P < 0.05$).

¹Values are represented as LSM ± maximum SEM.

²DT = disease treatment, P = parity class (1+2, ≥3), S = study (1a, 1b, 2, 3).

³LnVar = natural log-transformed variance, R_{auto} = lag-1 autocorrelation and MNR = mean negative residuals of milk yield, calculated over the period 10–56 DIM and 10–305 DIM. Skewness and MNR were transformed by a natural logarithm to obtain normally distributed residuals; back-transformed values are presented here.

related to health problems in early lactation, primarily during the first phase, with fewer relations observed later in lactation.

Lactation Curve Characteristics

The shape of the lactation curve, as represented by peak production, day of peak production, milk yield until 56 DIM, was related to health status and the presence or absence of a disease treatment in early lactation. In an earlier study, cows with one or more metabolic diseases

before 30 DIM had a later day of peak production, and cows with more than one metabolic disease before 30 DIM had a lower peak production and total milk yield in the first 60 DIM compared with healthy cows (Hostens et al., 2012). This was largely in line with our results, although some differences were observed between the studies, mainly concerning the relation between the number of diseases and the lactation curve characteristics. Cows with more than one metabolic disease before 30 DIM had a lower slope to peak production, a lower 305-d production, and a higher persistency compared with

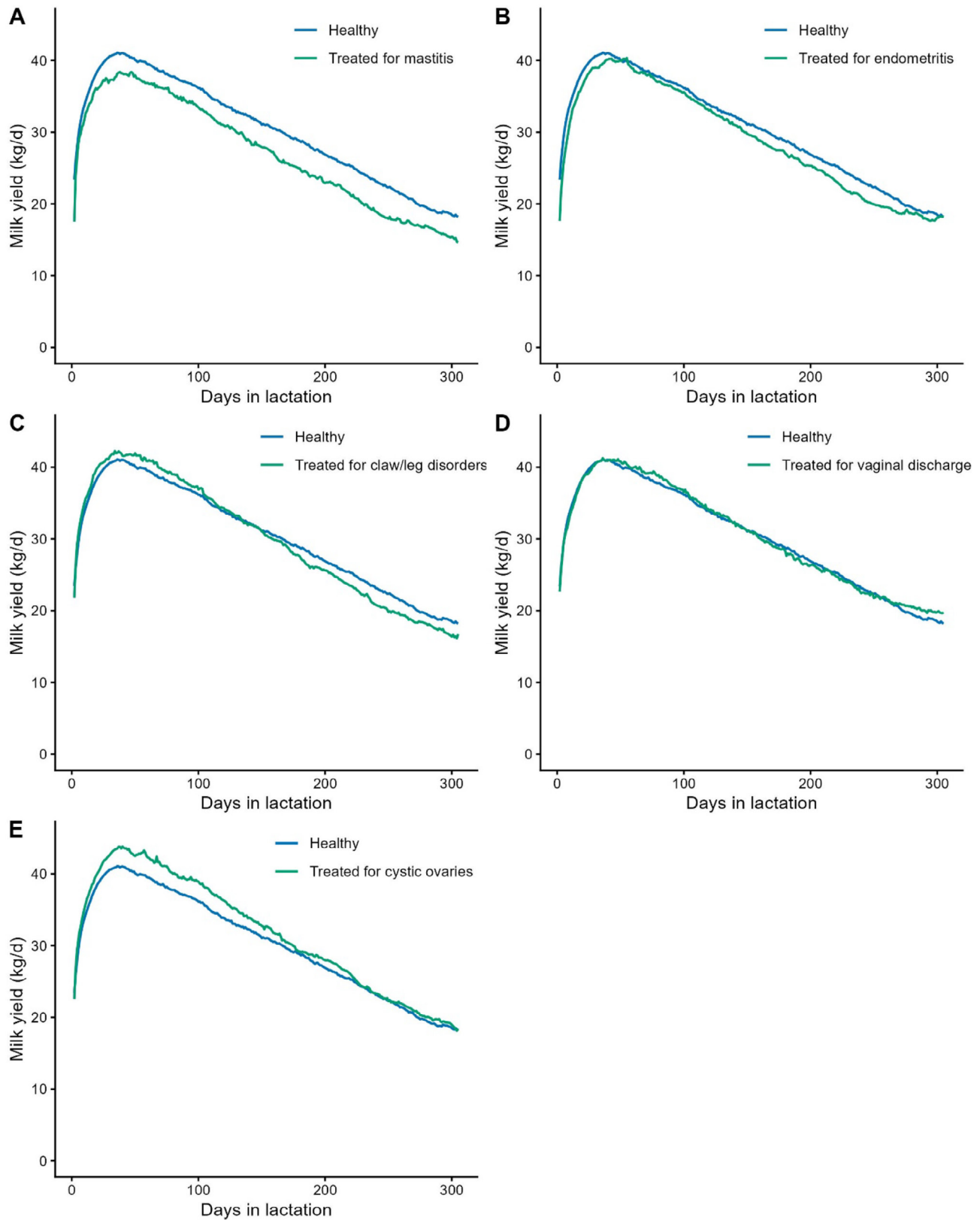


Figure 5. Lactation curves (kg of milk/d) until 305 DIM for cows without any disease treatment ($n = 222$) and cows treated for (A) mastitis ($n = 59$), (B) endometritis ($n = 65$), (C) claw and leg disorders ($n = 65$), (D) vaginal discharge ($n = 77$), and (E) cystic ovaries ($n = 62$) during the first 56 DIM.

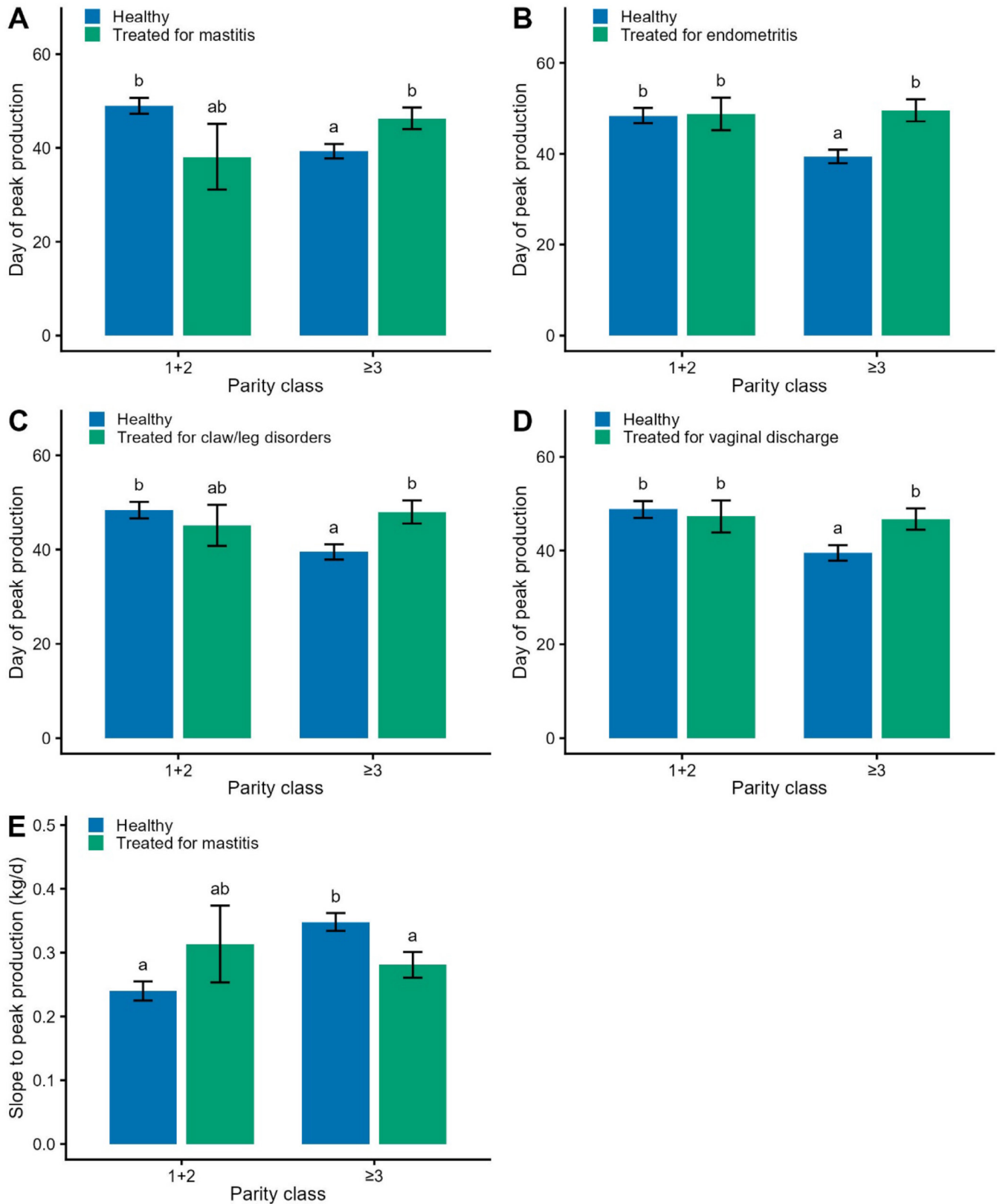


Figure 6. Interaction between disease treatment during 56 DIM and parity class (1+2, ≥ 3) for day of peak production for cows with (A) mastitis treatment, (B) endometritis treatment, (C) claw and leg treatments, (D) vaginal discharge treatment, and (E) slope to peak production for cows with mastitis treatment. Values represent LSM ± SE. Values with different letters (a,b) differ ($P < 0.05$).

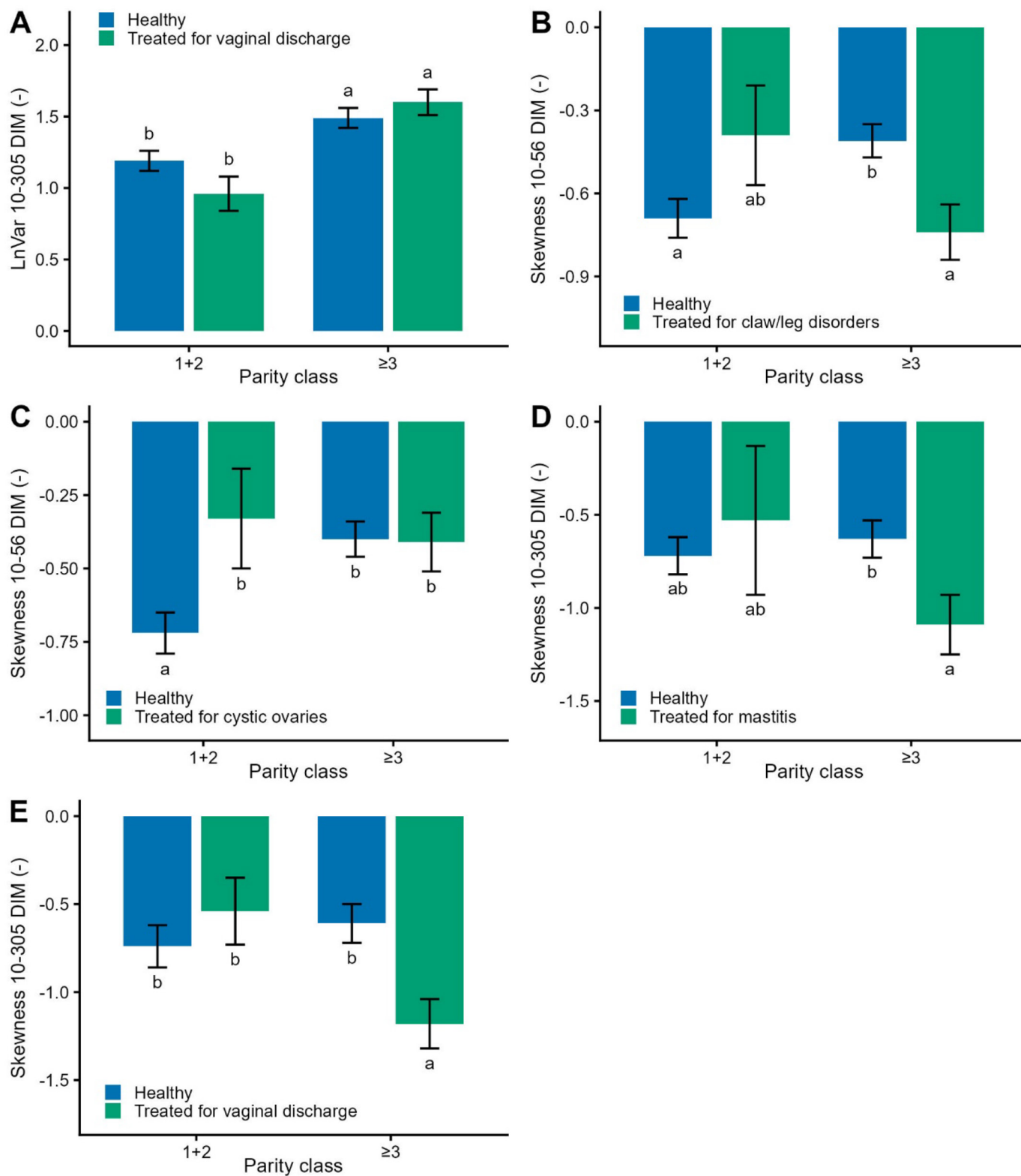


Figure 7. Interaction between disease treatment during 56 DIM and parity class (1+2, ≥ 3) for (A) LnVar during 10–305 DIM for treatment for cystic ovaries, skewness during 10–56 DIM for (B) claw and leg treatment, and (C) cystic ovaries, and skewness during 10–305 DIM for (D) mastitis treatment and (E) vaginal discharge treatment. Values represent LSM ± SE. Values with different letters (a,b) differ ($P < 0.05$).

healthy cows or cows with only one metabolic disease (Hostens et al., 2012), whereas the current study did not observe relations between the number of disease treatments and any of these lactation curve characteristics. We evaluated the number of any disease treatments before 56 DIM, whereas Hostens et al. (2012) only evaluated the number of metabolic diseases before 30 DIM. This may partly explain the different findings between the studies. Hence, the combination of the type and number of diseases seems to differently affect milk yield, and further research based on a bigger dataset is required to improve understanding of the relation between lactation curve characteristics and the type and number of diseases in early lactation.

The lactation curve for cows with systemic and inflammatory diseases in early lactation corresponded with the lactation curve for cows treated specifically for endometritis, but differed from cows treated for mastitis. Treatment for endometritis in early lactation was mainly related to the lactation curve characteristics in early lactation, whereas treatment for mastitis had a long-lasting effect by also reducing the 305-d production. Mastitis negatively affected milk yield (van Soest et al., 2016), with the largest impact around the mastitis case (Adriaens et al., 2021b; Wang et al., 2024), and when the case takes place in early lactation (Santos et al., 2004). In addition, clinical mastitis was related to a reduction of up to 11% in 305-d production and a suppression of milk yield throughout lactation (Hagnestam et al., 2007). It should be noted that repeated mastitis cases were reported to have a cumulative negative effect on milk yield (Costa et al., 2025). Mastitis cases after 56 DIM were not taken into account in this study, whereas cows with mastitis in early lactation are at an increased risk for mastitis later in lactation (Jamali et al., 2018), thereby potentially contributing to the reduced 305-d production in cows treated for mastitis. These findings indicate that health monitoring of mastitis may benefit from different or additional indicators that take into account the recurrent nature of mastitis. In addition, further research is required to determine whether other diseases also benefit from more disease-specific indicators.

Systemic and inflammatory diseases were negatively related to some milk yield characteristics in early lactation, whereas this relation was not found for cows with other health problems in early lactation. Other health problems in this study were mainly related to vaginal discharge, locomotion problems and cystic ovaries. These health problems were only marginally related to the lactation curve characteristics, as only a later day of peak production was seen for cows with claw and leg disorders or vaginal discharge. Cows with nonsystemic diseases likely experience less compromised energy intake, lipid mobilization, and metabolic stress compared with cows

with systemic diseases (Sordillo and Mavangira, 2014), thereby minimizing the negative relation with milk yield. In line with this, cows suffering from lameness changed their feeding behavior, characterized by fewer feeding visits but faster feeding rate and reduced feeding time compared with nonlame cows, whereas fresh matter intake or milk yield were not affected by the lameness cases (Thorup et al., 2016). However, severe lameness (score 4 out of 5) during lactation resulted in a 0.78-kg loss in daily milk yield compared with no lameness (score 1 out of 5) through lactation (Onyiro et al., 2008). The severity of the lameness case likely affects the change in feeding behavior, but when energy intake can be maintained by adjusted feeding behavior, the reduction in milk yield associated with lameness can be limited. Another study reported a negative impact of vaginal discharge on milk yield, but this also depended on the severity of the disease (Figueiredo et al., 2024). Cows with severe vaginal discharge in the first 2 wk postpartum had reduced milk yield until 300 DIM compared with cows with no or mild vaginal discharge, whereas milk yield was similar between cows with no and mild vaginal discharge (Figueiredo et al., 2024). The similar milk yield in cows with no or mild vaginal discharge is likely partly related to the absence of other disease symptoms in cows with mild vaginal discharge (Kawashima et al., 2018). Severe nonsystemic diseases seem to negatively affect milk yield, but this could not be evaluated in our study because only disease treatments, and not their severity, were available. Hence, disease severity seems important to take into account when evaluating the impact of diseases on milk yield characteristics in dairy cows.

Dynamic Indicators of Milk Yield

Cows with health problems during the first 56 DIM had a higher LnVar compared with healthy cows, whereas R_{auto} was not related to health status, but only with specific diseases. The LnVar was previously suggested to indicate the impact of disturbances, and R_{auto} was suggested to indicate the length of the impact of disturbances. A high LnVar and R_{auto} were suggested to be related to less resilient cows during this period leading to more health deviations, slower recovery, or both, compared with more resilient cows (Scheffer et al., 2018; Berghof et al., 2019). A decreased health status in early lactation is often related to a compromised energy metabolism and immunosuppression, thereby increasing susceptibility to secondary diseases in this period (Raboisson et al., 2014; Aleri et al., 2016). This decreased health status was mainly related to short-term perturbations in milk yield around the disease occurrence, indicated by the high LnVar, but the recovery capacity after the perturbation, indicated by the R_{auto} , was similar to healthy cows.

A slow recovery, indicated by an increased R_{auto} , was hypothesized in cows with a decreased health status in early lactation, but our results suggested that milk yield returned to the baseline quickly in these cows after disease treatment. This may be related to endocrine changes such as peripheral insulin resistance and downregulation of the hepatic growth hormone receptors (Lucy et al., 2001; Jiang et al., 2005) that facilitate the prioritization of milk yield under reduced glucose availability in early lactation. Quantifying short-term perturbations in milk yield, indicated by an increased LnVar, seems of added value in identifying cows with health problems in early lactation, as these cows could not be identified using lactation curve characteristics.

Although health status groups were not related to R_{auto} , cows with treatment for endometritis or cystic ovaries had a higher R_{auto} until 305 DIM compared with healthy cows. These treated cows failed to fully recover from the disorder in early lactation and experienced long-lasting deviations in their milk yield. This was not reflected in the lactation curve characteristics for cows treated for endometritis, but, interestingly, cows treated for cystic ovaries tended to have a higher 305-d production compared with healthy cows. This is likely because cystic ovaries are more prevalent in high-yielding cows, and high milk yield in early lactation has been identified as a risk factor for cystic ovaries later in lactation (López-Gatius et al., 2002). It may be that these high-yielding cows fail to recover from cystic ovaries throughout lactation, as a result of an persistent neuroendocrine imbalance in the hypothalamic-hypophyseal-ovarian axis (Garverick, 1997), thereby also inducing long-lasting deviations in milk yield. The R_{auto} seemed to provide important information about the long-lasting reduced recovery capacity of cows after treatment for endometritis or cystic ovaries, which could not be detected by using lactation curve characteristics.

A higher number of disease treatments until 56 DIM was related to increased LnVar from 10 to 56 DIM, irrespective of the treatment type, but relations with other dynamic indicators of milk yield were not present. Postpartum diseases are often intertwined (Sordillo and Mavangira, 2014), and comorbidity of postpartum diseases is high (Yáñez et al., 2022; do Nascimento et al., 2023). Cows with more deviations in blood serum values and clinical scores in the first 6 wk postpartum were considered to be less resilient, as these cows are no longer able to effectively adapt to all physiological and environmental changes in early lactation (van Dixhoorn et al., 2023). This is also reflected in our findings, as an increasing number of deviations in health status is related to more severe perturbations in milk yield.

Dynamic indicators of milk yield were previously related to health-related breeding values and clinical mastitis data. A lower full lactation LnVar was genetically

associated with better udder health, better longevity, and less ketosis (Poppe et al., 2020). Another study showed that cows suffering from mastitis in early lactation had a higher LnVar from 10 to 30 DIM and 10 to 305 DIM compared with healthy cows in early lactation (Kok et al., 2021). Our study did not report differences in LnVar between cows treated for mastitis and without any disease treatment in early lactation. Our study also included subclinical mastitis cases, which are typically not accompanied by systemic disease symptoms (Ashraf and Imran, 2020). This may explain the absence of relations between LnVar and mastitis in our study. Other results were in line with other studies, as cows treated for claw and leg disorders had a higher LnVar during 10–56 DIM and cows treated for cystic ovaries had a higher LnVar from 10 to 305 DIM compared with cows without any disease treatment. Based on this, it may be concluded that LnVar is related to both genotypic and clinical health traits of dairy cows. Studying the interaction between genotype and the incidence of a clinical health problems may even improve the value of dynamic indicators of milk yield in cow health monitoring. In addition, it is worthwhile further investigating the effect of different disease types on dynamic indicator of milk yield based on a bigger dataset as well as their specificity and sensitivity for different diseases. This can allow for more disease-specific application of the indicators in practice.

The MNR was another dynamic indicator of milk yield that was investigated in this study. No relations with health status group were present, but cows treated for claw and leg disorders in early lactation had a lower MNR in early lactation compared with cows without any disease treatment in this period. The MNR indicates the level of extreme negative deviations in milk yield, where a MNR close to zero indicates minimal extreme negative deviations from the expected curve (Poppe et al., 2022). Also in the current study, the MNR was indicative for cows treated for claw disorders, likely because cows with lameness showed drops in milk yield that took the longest to recover compared with other diseases. Moreover, our results indicate that the fluctuations in milk yield, including both positive and negative deviations indicated by the LnVar, are more relevant to determining health status than the drops in milk yield indicated by the MNR. This confirms the resilience theory of Scheffer et al. (2018), where the combination of the severity and frequency of the response to a challenge and the recovery after a challenge are important factors to consider to identify resilient cows. The MNR seems to be a relevant indicator for claw and leg disorders in the current study, whereas this was not the case for other diseases, but evaluation of results on a larger scale is recommended.

Skewness was investigated as dynamic indicator of milk yield, as a skewness around 0 was suggested to

indicate a good resilience, indicating a symmetric distribution of deviations from the individual lactation curve, caused by a limited response to disturbances (Scheffer et al., 2018; Berghof et al., 2019). In the current study, skewness was often affected by an interaction between health status group or disease treatments and parity class, but no clear results were found. The skewness measurement is highly sensitive to outliers or incorrect milk yield records and may therefore not be a suitable indicator, as also suggested by Poppe et al. (2020).

In the current study, health status in early lactation did not result in deviations in the lactation curve or dynamic indicator of milk yield later in lactation, but milk yield was mainly affected around the disease case. Another study reported a decrease in LnVar, but R_{auto} remained unchanged throughout lactation (Poppe et al., 2021). In addition, perturbations in milk yield until 138 DIM in higher parities were more severe with faster development rates, slower recovery, and higher milk losses compared with perturbations later in lactation (Adriaens et al., 2021a). The more severe perturbations in milk yield in early lactation may be partially explained by the fact that the incidence of diseases such as mastitis, hypocalcemia, and endometritis is highest in early lactation (until 30 DIM; Koeck et al., 2012; Wang et al., 2024), with a negative impact on milk yield (Goff, 2008; Husnain et al., 2023; Sguizzato et al., 2024). The current study does not show long-term effects of a decreased health status in early lactation on milk yield dynamics later in lactation.

The variation in milk yield, indicated by LnVar and MNR until 305 DIM, was higher for cows of parity ≥ 3 compared with cows of parity 1 and 2. Others have also reported an increase in LnVar (Kok et al., 2021; Poppe et al., 2021) or an increased number of perturbations in milk yield (Adriaens et al., 2021a) with increasing parity. This is likely related to the higher risk for diseases in older cows (Lee and Kim, 2006), and due to higher milk yield in older cows than younger cows (Kok et al., 2021).

Practical Implications and Perspectives

Data from 4 experimental studies were used for the current study. Cows were subjected to different DPL or different VWP until first insemination in these studies. It is important to note that only healthy cows were selected to enter these experiments, which may have biased the results. Disease incidence in early lactation was lower compared with the study of Koeck et al. (2012), but similar to Ribeiro et al. (2013). Differences in milk yield among studies were present, but this had a limited effect on the results, as the interaction for health status and study was never significant. Although the study and treatment effects are corrected for in the analysis, evaluation

of the results on a wide range of commercial farms is recommended.

Next to the commonly used lactation curve characteristics, dynamic indicators of milk yield seem to be of added value as indicators for disease, as they identified cows with other health problems and different numbers of disease treatments, whereas this was not the case for lactation curve characteristics. The fluctuations in milk yield are better captured in the dynamic indicators of milk yield compared with using lactation curve characteristics, thereby taking more information into account about the response to, and recovery after, a challenge (Scheffer et al., 2018). This possibly contributes to the clarification that cows with other health problems deviate from the healthy cows for the dynamic indicators of milk yield, but not for the lactation curve characteristics. Further investigation of the predictive value of these indicators, including their specificity and sensitivity, would enhance practical application in health monitoring tools, especially when tailored to specific diseases. It is recommended to use disease diagnoses, instead of disease treatments, as read outs for health status in future studies to avoid a large effect of differences in treatment protocols and disease severity on study results. Integration of milk yield dynamic indicators with other data, such as fluctuations in milk composition (Ehsaninia et al., 2019), SCC (de Haas et al., 2008), or behavioral patterns (van Dixhoorn et al., 2018, 2023), would be worthwhile to explore. In addition, fluctuations in milk yield are heritable (Elgersma et al., 2018), and further investigation of the predictive value of within-cow variability in milk yield dynamics may be valuable in optimizing health management for individual cows.

CONCLUSIONS

The shape of the lactation curve and the fluctuations in milk yield during the first 56 DIM were related to health status during the first 56 DIM. Although lactation curve characteristics specifically indicated cows with a systemic or inflammatory health problem, dynamic indicators of milk yield indicated both cows with a systemic or inflammatory health problem and cows with other health problems. Peak production and total milk yield until 56 DIM were lower in cows with systemic or inflammatory problems during the first 56 DIM compared with cows with no or other health problems. This indicated that cows with health problems in early lactation showed more perturbations in milk yield than healthy cows, and the response seemed to be disease-dependent. The impact of specific diseases and the number of disease treatments on milk yield dynamics should be further investigated to potentially allow for application in on-farm health monitoring tools.

NOTES

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Nonstandard abbreviations used: DPL = dry period length; FPCM = fat- and protein-corrected milk yield; HS = health status group; LnVar = natural log-transformed variance; MNR = mean of negative residuals; MY = milk yield; NHP = no health problems; OHP = other health problems; P = parity class; R_{auto} = lag-1 autocorrelation; S = study; SHP = systemic or inflammatory health problems; VWP = voluntary waiting period.

REFERENCES

- Adriaens, I., I. van den Brulle, L. D’Anvers, J. M. E. Statham, K. Geerinckx, S. De Vliegheer, S. Piepers, and B. Aernouts. 2021a. Milk losses and dynamics during perturbations in dairy cows differ with parity and lactation stage. *J. Dairy Sci.* 104:405–418. <https://doi.org/10.3168/jds.2020-19195>.
- Adriaens, I., I. Van Den Brulle, K. Geerinckx, L. D’Anvers, S. De Vliegheer, and B. Aernouts. 2021b. Milk losses linked to mastitis treatments at dairy farms with automatic milking systems. *Prev. Vet. Med.* 194:105420. <https://doi.org/10.1016/j.prevetmed.2021.105420>.
- Aleri, J. W., B. C. Hine, M. F. Pyman, P. D. Mansell, W. J. Wales, B. Mallard, and A. D. Fisher. 2016. Periparturient immunosuppression and strategies to improve dairy cow health during the periparturient period. *Res. Vet. Sci.* 108:8–17. <https://doi.org/10.1016/j.rvsc.2016.07.007>.
- Ashraf, A., and M. Imran. 2020. Causes, types, etiological agents, prevalence, diagnosis, treatment, prevention, effects on human health and future aspects of bovine mastitis. *Anim. Health Res. Rev.* 21:36–49. <https://doi.org/10.1017/S1466252319000094>.
- Ben Abdelkrim, A., L. Puillet, P. Gomes, and O. Martin. 2021. Lactation curve model with explicit representation of perturbations as a phenotyping tool for dairy livestock precision farming. *Animal* 15:100074. <https://doi.org/10.1016/j.animal.2020.100074>.
- Berghof, T. V. L., M. Poppe, and H. A. Mulder. 2019. Opportunities to improve resilience in animal breeding programs. *Front. Genet.* 9:692.
- Burgers, E. E. A., R. M. A. Goselink, R. M. Bruckmaier, J. J. Gross, R. Jorritsma, B. Kemp, A. Kok, and A. T. van Kneegsel. 2023. Effect of voluntary waiting period on metabolism of dairy cows during different phases of the lactation. *Journal of Animal Science* 101(skad194).
- Burgers, E. E. A., A. Kok, R. M. A. Goselink, H. Hogeveen, B. Kemp, and A. T. M. van Kneegsel. 2021. Effects of extended voluntary waiting period from calving until first insemination on body condition, milk yield, and lactation persistency. *J. Dairy Sci.* 104:8009–8022. <https://doi.org/10.3168/jds.2020-19914>.
- Carvalho, M. R., F. Peñagaricano, J. E. P. Santos, T. J. DeVries, B. W. McBride, and E. S. Ribeiro. 2019. Long-term effects of postpartum clinical disease on milk production, reproduction, and culling of dairy cows. *J. Dairy Sci.* 102:11701–11717. <https://doi.org/10.3168/jds.2019-17025>.
- Chen, J., J. J. Gross, H. A. van Dorland, G. J. Remmelink, R. M. Bruckmaier, B. Kemp, and A. T. M. van Kneegsel. 2015. Effects of dry period length and dietary energy source on metabolic status and hepatic gene expression of dairy cows in early lactation. *J. Dairy Sci.* 98:1033–1045. <https://doi.org/10.3168/jds.2014-8612>.
- Costa, A., H. Bovenhuis, C. Egger-Danner, B. Fuerst-Waltl, M. Bou-tinaud, J. Guinard-Flament, W. Obritzhauser, G. Visentin, and M. Penasa. 2025. Mastitis has a cumulative and lasting effect on milk yield and lactose content in dairy cows. *J. Dairy Sci.* 108:635–650. <https://doi.org/10.3168/jds.2024-25467>.
- de Haas, Y., W. Ouweltjes, J. Napel, J. J. Windig, and G. de Jong. 2008. Alternative somatic cell count traits as mastitis indicators for genetic selection. *J. Dairy Sci.* 91:2501–2511. <https://doi.org/10.3168/jds.2007-0459>.
- De Vliegheer, S., H. W. Barkema, H. Stryhn, G. Opsomer, and A. de Kruif. 2005. Impact of early lactation somatic cell count in heifers on milk yield over the first lactation. *J. Dairy Sci.* 88:938–947. [https://doi.org/10.3168/jds.S0022-0302\(05\)72761-2](https://doi.org/10.3168/jds.S0022-0302(05)72761-2).
- do Nascimento, A. M. M., C. M. S. de Souza, A. C. D. Oliveira, M. G. Blagitz, E. M. Ramos Sanchez, A. M. M. P. Della Libera, R. M. H. Leite, A. C. C. Fernandes, and F. N. Souza. 2023. The bovine leukemia virus infection prolongs immunosuppression in dairy cows during the periparturient period by sustaining higher expression of immunological checkpoints in T cells. *Vet. Immunol. Immunopathol.* 263:110636.
- Dubuc, J., T. F. Duffield, K. E. Leslie, J. S. Walton, and S. J. LeBlanc. 2011. Effects of postpartum uterine diseases on milk production and culling in dairy cows. *J. Dairy Sci.* 94:1339–1346. <https://doi.org/10.3168/jds.2010-3758>.
- Ehsaninia, J., N. Ghavi Hosseini-Zadeh, and A. A. Shadparvar. 2019. Estimation of genetic variation for macro- and micro-environmental sensitivities of milk yield and composition in Holstein cows using double hierarchical generalized linear models. *J. Dairy Res.* 86:145–153. <https://doi.org/10.1017/S0022029919000293>.
- Elgersma, G. G., G. de Jong, R. van der Linde, and H. A. Mulder. 2018. Fluctuations in milk yield are heritable and can be used as a resilience indicator to breed healthy cows. *J. Dairy Sci.* 101:1240–1250. <https://doi.org/10.3168/jds.2017-13270>.
- Figueiredo, C. C., S. Casaro, F. Cunha, V. R. Merenda, E. B. de Oliveira, P. Pinedo, J. E. P. Santos, R. C. Chebel, G. M. Schuenemann, R. C. Bicalho, R. O. Gilbert, S. R. Zas, C. M. Seabury, G. Rosa, W. W. Thatcher, R. S. Bisinotto, and K. N. Galvão. 2024. Evaluating differences in milk production, reproductive performance, and sur-

- vival associated with vaginal discharge characteristics and fever in postpartum dairy cows. *J. Dairy Sci.* 107:6079–6089. <https://doi.org/10.3168/jds.2023-23905>.
- Garverick, H. A. 1997. Ovarian follicular cysts in dairy cows. *J. Dairy Sci.* 80:995–1004. [https://doi.org/10.3168/jds.S0022-0302\(97\)76025-9](https://doi.org/10.3168/jds.S0022-0302(97)76025-9).
- Goff, J. P. 2008. The monitoring, prevention, and treatment of milk fever and subclinical hypocalcemia in dairy cows. *Vet. J.* 176:50–57. <https://doi.org/10.1016/j.tvjl.2007.12.020>.
- Goff, J. P., and R. L. Horst. 1997. Physiological changes at parturition and their relationship to metabolic disorders. *J. Dairy Sci.* 80:1260–1268. [https://doi.org/10.3168/jds.S0022-0302\(97\)76055-7](https://doi.org/10.3168/jds.S0022-0302(97)76055-7).
- Gröhn, Y. T., D. J. Wilson, R. N. González, J. A. Hertl, H. Schulte, G. Bennett, and Y. H. Schukken. 2004. Effect of pathogen-specific clinical mastitis on milk yield in dairy cows. *J. Dairy Sci.* 87:3358–3374. [https://doi.org/10.3168/jds.S0022-0302\(04\)73472-4](https://doi.org/10.3168/jds.S0022-0302(04)73472-4).
- Hagnestam, C., U. Emanuelson, and B. Berglund. 2007. Yield losses associated with clinical mastitis occurring in different weeks of lactation. *J. Dairy Sci.* 90:2260–2270. <https://doi.org/10.3168/jds.2006-583>.
- Hostens, M., J. Ehrlich, B. Van Ranst, and G. Opsomer. 2012. On-farm evaluation of the effect of metabolic diseases on the shape of the lactation curve in dairy cows through the MilkBot lactation model. *J. Dairy Sci.* 95:2988–3007. <https://doi.org/10.3168/jds.2011-4791>.
- Husnain, A., U. Arshad, M. B. Poindexter, R. Zimpel, M. N. Marinho, M. C. Perdomo, P. Fan, K. C. Jeong, C. D. Nelson, I. M. Sheldon, J. J. Bromfield, and J. E. P. Santos. 2023. Induced endometritis in early lactation compromises production and reproduction in dairy cows. *J. Dairy Sci.* 106:4198–4213. <https://doi.org/10.3168/jds.2022-22846>.
- Jamali, H., H. W. Barkema, M. Jacques, E.-M. Lavallée-Bourget, F. Malouin, V. Saini, H. Stryhn, and S. Dufour. 2018. Invited review: Incidence, risk factors, and effects of clinical mastitis recurrence in dairy cows. *J. Dairy Sci.* 101:4729–4746. <https://doi.org/10.3168/jds.2017-13730>.
- Jiang, H., M. C. Lucy, B. A. Crooker, and W. E. Beal. 2005. Expression of Growth Hormone Receptor 1A mRNA is Decreased in Dairy Cows but not in Beef Cows at Parturition. *J. Dairy Sci.* 88:1370–1377. [https://doi.org/10.3168/jds.S0022-0302\(05\)72804-6](https://doi.org/10.3168/jds.S0022-0302(05)72804-6).
- Kawashima, C., M. Suwanai, T. Honda, M. Teramura, K. Kida, M. Hanada, A. Miyamoto, and M. Matsui. 2018. Relationship of vaginal discharge characteristics evaluated by Metricheck device to metabolic status in postpartum dairy cows. *Reprod. Domest. Anim.* 53:1396–1404. <https://doi.org/10.1111/rda.13255>.
- Keßler, F., R. Wellmann, M. G. G. Chagunda, and J. Bennewitz. 2025. Toward a resilience selection index with indicator traits in German Holstein dairy cattle. *J. Dairy Sci.* 108:726–734. <https://doi.org/10.3168/jds.2024-25323>.
- Koeck, A., F. Miglior, D. F. Kelton, and F. S. Schenkel. 2012. Health recording in Canadian Holsteins: Data and genetic parameters. *J. Dairy Sci.* 95:4099–4108. <https://doi.org/10.3168/jds.2011-5127>.
- Koenker, R. 2009. quantreg: Quantile regression. Accessed Oct. 15, 2025. <http://CRAN.R-project.org/package=quantreg>.
- Kok, A., G. Tsousis, G. Niozas, B. Kemp, M. Kaske, and A. T. M. van Kneegsel. 2021. Short communication: Variance and autocorrelation of deviations in daily milk yield are related with clinical mastitis in dairy cows. *Animal* 15:100363. <https://doi.org/10.1016/j.animal.2021.100363>.
- LeBlanc, S. 2010. Monitoring metabolic health of dairy cattle in the transition period. *J. Reprod. Dev.* 56(Suppl):S29–S35. <https://doi.org/10.1262/jrd.1056S29>.
- Lee, J.-Y., and I.-H. Kim. 2006. Advancing parity is associated with high milk production at the cost of body condition and increased periparturient disorders in dairy herds. *J. Vet. Sci.* 7:161. <https://doi.org/10.4142/jvs.2006.7.2.161>.
- López-Gatius, F., P. Santolaria, J. Yáñez, M. Fenech, and M. López-Béjar. 2002. Risk factors for postpartum ovarian cysts and their spontaneous recovery or persistence in lactating dairy cows. *Theriogenology* 58:1623–1632. [https://doi.org/10.1016/S0093-691X\(02\)01046-4](https://doi.org/10.1016/S0093-691X(02)01046-4).
- Lucy, M. C., H. Jiang, and Y. Kobayashi. 2001. Changes in the somatotrophic axis associated with the initiation of lactation. *J. Dairy Sci.* 84:E113–E119. [https://doi.org/10.3168/jds.S0022-0302\(01\)70205-6](https://doi.org/10.3168/jds.S0022-0302(01)70205-6).
- Onyiro, O. M., J. Offer, and S. Brotherstone. 2008. Risk factors and milk yield losses associated with lameness in Holstein-Friesian dairy cattle. *Animal* 2:1230–1237. <https://doi.org/10.1017/S1751731108002279>.
- Pérez-Báez, J., T. V. Silva, C. A. Risco, R. C. Chebel, F. Cunha, A. De Vries, J. E. P. Santos, F. S. Lima, P. Pinedo, G. M. Schuenemann, R. C. Bicalho, R. O. Gilbert, S. Rodríguez-Zas, C. M. Seabury, G. Rosa, W. W. Thatcher, and K. N. Galvão. 2021. The economic cost of metritis in dairy herds. *J. Dairy Sci.* 104:3158–3168. <https://doi.org/10.3168/jds.2020-19125>.
- Poppe, M., G. Bonekamp, M. L. van Pelt, and H. A. Mulder. 2021. Genetic analysis of resilience indicators based on milk yield records in different lactations and at different lactation stages. *J. Dairy Sci.* 104:1967–1981. <https://doi.org/10.3168/jds.2020-19245>.
- Poppe, M., H. A. Mulder, M. L. van Pelt, E. Mullaart, H. Hogeveen, and R. F. Veerkamp. 2022. Development of resilience indicator traits based on daily step count data for dairy cattle breeding. *Genet. Sel. Evol.* 54:21. <https://doi.org/10.1186/s12711-022-00713-x>.
- Poppe, M., R. F. Veerkamp, M. L. van Pelt, H. A. Mulder. 2020. Exploration of variance, autocorrelation, and skewness of deviations from lactation curves as resilience indicators for breeding. *J. Dairy Sci.* 103:1667–1684. <https://doi.org/10.3168/jds.2019-17290>.
- Puerto, M. A., E. Shepley, R. I. Cue, D. Warner, J. Dubuc, and E. Vasseur. 2021. The hidden cost of disease: I. Impact of the first incidence of mastitis on production and economic indicators of primiparous dairy cows. *J. Dairy Sci.* 104:7932–7943. <https://doi.org/10.3168/jds.2020-19584>.
- Raboisson, D., M. Mounié, and E. Maigné. 2014. Diseases, reproductive performance, and changes in milk production associated with sub-clinical ketosis in dairy cows: A meta-analysis and review. *J. Dairy Sci.* 97:7547–7563. <https://doi.org/10.3168/jds.2014-8237>.
- Randall, L. V., M. J. Green, M. G. G. Chagunda, C. Mason, L. E. Green, and J. N. Huxley. 2016. Lameness in dairy heifers; impacts of hoof lesions present around first calving on future lameness, milk yield and culling risk. *Prev. Vet. Med.* 133:52–63. <https://doi.org/10.1016/j.prevetmed.2016.09.006>.
- Ribeiro, E. S., and M. R. Carvalho. 2017. Impact and mechanisms of inflammatory diseases on embryonic development and fertility in cattle. *Anim. Reprod.* 14:589–600. <https://doi.org/10.21451/1984-3143-AR1002>.
- Ribeiro, E. S., F. S. Lima, L. F. Greco, R. S. Bisinotto, A. P. Monteiro, M. Favoreto, H. Ayres, R. S. Marsola, N. Martinez, W. W. Thatcher, and J. E. Santos. 2013. Prevalence of periparturient diseases and effects on fertility of seasonally calving grazing dairy cows supplemented with concentrates. *J. Dairy Sci.* 96:5682–5697. <https://doi.org/10.3168/jds.2012-6335>.
- Santos, J. E. P., R. L. A. Cerri, M. A. Ballou, G. E. Higginbotham, and J. H. Kirk. 2004. Effect of timing of first clinical mastitis occurrence on lactational and reproductive performance of Holstein dairy cows. *Anim. Reprod. Sci.* 80:31–45. [https://doi.org/10.1016/S0378-4320\(03\)00133-7](https://doi.org/10.1016/S0378-4320(03)00133-7).
- Scheffer, M., J. E. Bolhuis, D. Borsboom, T. G. Buchman, S. M. W. Gijzel, D. Goulson, J. E. Kammenga, B. Kemp, I. A. van de Leemput, S. Levin, C. M. Martin, R. J. F. Melis, E. H. van Nes, L. M. Romero, and M. G. M. Olde Rikkert. 2018. Quantifying resilience of humans and other animals. *Proc. Natl. Acad. Sci. USA* 115:11883–11890. <https://doi.org/10.1073/pnas.1810630115>.
- Sguizzato, A. L. L., T. E. da Silva, J. C. C. Chagas, A. M. Argüelo, N. M. Gonçalves, and M. I. Marcondes. 2024. Understanding the dynamics of mastitis in milk yield: Decoding onset and recovery patterns in response to mastitis occurrence. *JDS Commun.* 5:669–673. <https://doi.org/10.3168/jdsc.2024-0579>.
- Sordillo, L. M., and V. Mavangira. 2014. The nexus between nutrient metabolism, oxidative stress and inflammation in transition cows. *Anim. Prod. Sci.* 54:1204–1214. <https://doi.org/10.1071/AN14503>.
- Sumner, C. L., M. A. G. von Keyserlingk, and D. M. Weary. 2018. Perspectives of farmers and veterinarians concerning dairy cattle welfare. *Anim. Front.* 8:8–13. <https://doi.org/10.1093/af/vfx006>.
- Sundrum, A. 2015. Metabolic disorders in the transition period indicate that the dairy cows' ability to adapt is overstressed. *Animals (Basel)* 5:978–1020. <https://doi.org/10.3390/ani5040395>.
- Thorup, V. M., B. L. Nielsen, P.-E. Robert, S. Giger-Reverdin, J. Konka, C. Michie, and N. C. Friggens. 2016. Lameness affects cow feed-

- ing but not rumination behavior as characterized from sensor data. *Front. Vet. Sci.* 3:37. <https://doi.org/10.3389/fvets.2016.00037>.
- van Dixhoorn, I. D. E., R. M. de Mol, S. K. Schnabel, J. T. N. van der Werf, S. van Mourik, J. E. Bolhuis, J. M. J. Rebel, and C. G. van Reenen. 2023. Behavioral patterns as indicators of resilience after parturition in dairy cows. *J. Dairy Sci.* 106:6444–6463. <https://doi.org/10.3168/jds.2022-22891>.
- van Dixhoorn, I. D. E., R. M. de Mol, J. T. N. van der Werf, S. van Mourik, and C. G. van Reenen. 2018. Indicators of resilience during the transition period in dairy cows: A case study. *J. Dairy Sci.* 101:10271–10282. <https://doi.org/10.3168/jds.2018-14779>.
- van Hoeij, R. J., J. Dijkstra, R. M. Bruckmaier, J. J. Gross, T. Lam, G. J. Remmelink, B. Kemp, and A. T. M. van Knegsel. 2017. The effect of dry period length and postpartum level of concentrate on milk production, energy balance, and plasma metabolites of dairy cows across the dry period and in early lactation. *J. Dairy Sci.* 100:5863–5879. <https://doi.org/10.3168/jds.2016-11703>.
- van Knegsel, A. T. M., G. J. Remmelink, S. Jorjeng, V. Fievez, and B. Kemp. 2014. Effect of dry period length and dietary energy source on energy balance, milk yield, and milk composition of dairy cows. *J. Dairy Sci.* 97:1499–1512. <https://doi.org/10.3168/jds.2013-7391>.
- van Soest, F. J. S., I. M. G. A. Santman-Berends, T. J. G. M. Lam, and H. Hogeveen. 2016. Failure and preventive costs of mastitis on Dutch dairy farms. *J. Dairy Sci.* 99:8365–8374. <https://doi.org/10.3168/jds.2015-10561>.
- Venjakob, P. L., L. Pieper, W. Heuwieser, and S. Borchardt. 2018. Association of postpartum hypocalcemia with early-lactation milk yield, reproductive performance, and culling in dairy cows. *J. Dairy Sci.* 101:9396–9405. <https://doi.org/10.3168/jds.2017-14202>.
- Wang, A., G. Su, L. F. Brito, H. Zhang, R. Shi, D. Liu, G. Guo, and Y. Wang. 2024. Investigating the relationship between fluctuations in daily milk yield as resilience indicators and health traits in Holstein cattle. *J. Dairy Sci.* 107:1535–1548. <https://doi.org/10.3168/jds.2023-23495>.
- Wankhade, P. R., A. Manimaran, A. Kumaresan, S. Jeyakumar, K. P. Ramesha, V. Sejian, D. Rajendran, and M. R. Varghese. 2017. Metabolic and immunological changes in transition dairy cows: A review. *Vet. World* 10:1367–1377. <https://doi.org/10.14202/vetworld.2017.1367-1377>.
- Yáñez, U., P. G. Herradón, J. J. Becerra, A. I. Peña, and L. A. Quintela. 2022. Relationship between postpartum metabolic status and sub-clinical endometritis in dairy cattle. *Animals (Basel)* 12:242.

ORCIDS

- B. G. C. de Bruijn,  <https://orcid.org/0000-0003-3705-9316>
 I. D. E. van Dixhoorn,  <https://orcid.org/0000-0003-3642-2523>
 B. Kemp,  <https://orcid.org/0000-0002-9765-9105>
 A. T. M. van Knegsel  <https://orcid.org/0000-0003-1959-3363>